

Project Proposal

Project Title

Machine Learning-Based Price Prediction and Analysis of Airbnb Listings Across European Cities

Dataset

- **Name:** Airbnb Market Data (Europe)
- **Source:** Kaggle Airbnb Dataset (Europe listings & rates)
- **Link:** <https://www.kaggle.com/datasets/jasonairroi/airbnb-market-data-europe>
- **Size:**
 - Listings dataset: 95898 rows × 63 columns
 - Rates dataset: 1115174 rows × 19 columns

Target Variable

- `booked_rate_avg` (average booked price per listing)

Type of Task

- **Regression** (continuous value prediction)

Problem Statement

Airbnb pricing is influenced by multiple factors such as location, property characteristics, and demand patterns. However, determining optimal pricing remains challenging for hosts due to market complexity and data variability.

This project aims to develop and evaluate machine learning models capable of predicting Airbnb listing prices across European cities, while also identifying key drivers of pricing and assessing model robustness and practical applicability.

Research Questions

RQ1 — Baseline Performance

How accurately can baseline supervised learning models predict Airbnb listing prices across European cities?

RQ2 — Model Comparison

Which supervised learning model achieves the best predictive performance for Airbnb listing price prediction, and how does it compare to alternative models?

RQ3 — Effect of Preprocessing

How do different data preprocessing techniques (imputation, encoding, scaling, and outlier handling) influence model performance?

RQ4 — Feature Importance and Interpretability

Which listing characteristics most strongly influence Airbnb pricing?

RQ5 — Sensitivity to Evaluation Metrics

How does model ranking change when evaluated using different regression metrics (MAE, RMSE, R^2)?

RQ6 — Robustness and Generalization

How robust is the selected model under different validation strategies and data perturbations?

RQ7 — Practical Usefulness and Final Recommendation

To what extent can machine learning models support data-driven pricing strategies for Airbnb hosts?

Proposed Methodology

1. **Data Loading and Cleaning**
 - Handle missing values
 - Convert price-related variables to numeric format
 - Remove invalid or extreme values
2. **Feature Selection**
 - Remove ID, URL, and high-cardinality text columns
 - Avoid leakage by excluding other price-related features
3. **Data Preprocessing**
 - Imputation (median/mode)
 - Encoding categorical variables
 - Feature scaling (where necessary)
 - Outlier handling
4. **Model Training**
 - Train-test split (80/20)
 - Cross-validation
5. **Model Evaluation**
 - Compare performance across models and preprocessing strategies
6. **Interpretability Analysis**
 - Feature importance extraction
 - Model explanation techniques

Machine Learning Models

- Linear Regression (baseline)
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting / XGBoost
- Support Vector Regression (SVR)

Evaluation Metrics

- **MAE (Mean Absolute Error)** → interpretability
- **RMSE (Root Mean Squared Error)** → penalizes large errors
- **R² Score** → explained variance

Expected Figures and Tables

Figures

- Distribution of target variable
- Model performance comparison (bar charts)
- Feature importance plots
- Residual plots
- Cross-validation performance plots

Tables

- Summary statistics of dataset
- Model performance comparison table
- Preprocessing impact comparison
- Metric sensitivity comparison